

Sahit Chintalapudi

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Education

Massachusetts Institute of Technology

Ph.D Candidate, EECS, [S.M. 2024]

Advised by Dr. Leslie Pack Kaelbling and Dr. Tomás Lozano-Pérez

Boston

2020–Present

Georgia Institute of Technology

B.S Computer Science, GPA: 3.94

Concentrations in Intelligence and Theory

Atlanta

2016–2019

Publications

Sahit Chintalapudi, Leslie Kaelbling, and Tomás Lozano-Pérez. Bi-level belief space search for compliant part mating under uncertainty. *Workshop on the Algorithmic Foundations of Robotics (WAFR)*, 2024.

Keshav Kolar*, Sahit Chintalapudi*, Byron Boots, and Mustafa Mukadam. Online motion planning over multiple homotopy classes with gaussian process inference. *Proceedings of the International Conference on Intelligent Robots and Systems (IROS)*, 2019.

Vinitha Ranganeni, Sahit Chintalapudi, Oren Salzman, and Maxim Likhachev. Effective footstep planning using homotopy-class guidance. *Artificial Intelligence*, 286:103346, 2020.

William Shen, Nishanth Kumar, Sahit Chintalapudi, Jie Wang, Christopher Watson, Edward Hu, Jing Cao, Dinesh Jayaraman, Leslie Pack Kaelbling, and Tomás Lozano-Pérez. Tiptop: A modular open-vocabulary planning system for robotic manipulation. *arXiv preprint arXiv:2603.09971*, 2026.

Maggie Huili Yao, Sahit Chintalapudi, Akash Anand, Megha Tippur, Tomás Lozano-Pérez, Leslie Pack Kaelbling, and Aditya Agarwal. Proxy: Learning object-centric manipulation policies from a single rgb-d image. In *RSS 2026 Workshop on Diffusion for Robot Learning*.

Research Experience

Autodesk

Robot Learning Intern

San Francisco

May 2025 2020–August 2025

- o Trained PPO (online) and Decision Transformer (offline) policies for 100 low-clearance robotic assembly tasks.
- o Studied how Mixture-of-Expert layers affect generalization in 10M-parameter Decision Transformers.
- o Deployed Decision Transformer policies trained in IsaacLab on a UR-10e robot via sim-to-real transfer.

DeepMind

Research Engineering Intern

London

March 2020–August 2020

- o Experimented with curricula generation methods for RL agents in the context of autonomous stacking.
- o Developed internal infrastructure for collecting human demonstrations of manipulation in simulated environments.

Georgia Institute of Technology: Robot Learning Lab**Atlanta***Undergraduate Research Assistant, advised by Dr. Byron Boots**2017–2019*

- o Used C++, MATLAB, and the GTSAM toolbox to model the planning problem with a factor graph that adapted in real time to environment changes. In an environment with randomly moving obstacles, this algorithm reduced collision intensity by at least 37% compared to other approaches.
- o Extending existing implementations of Model Predictive Control algorithms to run on the AutoRally platform and log data for Value Function Approximation.

University of Washington: Human-Centered Robotics Lab**Seattle***Undergraduate Research Assistant, advised by Dr. Maya Cakmak**May 2019–Aug 2019*

- o Implemented a Jacobian-Based Full-Body Controller for the Fetch Robot to perform research on mobile manipulation with ROS and C++.
- o Researched Task Decomposition to facilitate high dimensional planning in the context of autonomous cleaning.

Carnegie Mellon University: Searched Based Planning Lab**Pittsburgh***Robotics Institute Summer Scholar, advised by Dr. Maxim Likhachev**June 2018–Aug 2018*

- o Developed C++ Software for a humanoid footstep planner which plans 16-128 times faster than the baseline approach in environments with many obstacles

Skills

Languages: Python, C++, MATLAB, Java**Tools:** Drake, ROS, Tensorflow, PyTorch, NumPy, Linux, L^AT_EX, MuJoCo, Eigen, GTSAM

Awards

2019 President's Undergraduate Research Award: Georgia Tech Grant**1st Place:** Google Tech Challenge 2019**Best Collegiate Speed Demons Team:** Sparkfun Autonomous Vehicle Competition 2018

Projects

Transformer Policies that Reason Under Uncertainty*Machine Learning Final Project**Fall 2021*

- o Trained a transformer to perform decision making for long-horizon tasks with discrete action spaces
- o Used label smoothing and temperature scaling to improve model calibration

Control with Sums-Of-Squares Polynomials for Hybrid Autonomous Systems*Underactuated Robotics Final Project**Spring 2021*

- o Solved a relaxation of the HJB equation for a multimodal cartpole variant

MeleeML*Interactive Robot Learning Final Project**Fall 2019*

- o Trained a Generative Adversarial Imitation Learning (GAIL) agent to learn how to play Super Smash Brothers Melee (SSBM) from Human Demonstration using PyTorch
- o Designed and implemented an advantage actor-critic model to play SSBM by training against CPUs

RoboRacing: Autonomous RC Vehicles*Software Lead**2016–2018*

- o Prototyped deep network architectures in Keras for learning steering angles from images of the road.
- o Developed and Tuned the plant PID controller on an Arduino

Relevant Coursework

MIT: Algorithms for Inference, Underactuated Robotics, Optimization Methods, Machine Learning

Georgia Tech: Interactive Robot Learning (Graduate Course), Computer Vision, Honors Probability and Statistics, Robotics and Perception, Machine Learning

Service

Reviewer: T-RO, RA-L, CoRL, ICRA

Graduate Application Assistance Program Mentor 2020-2023: Mentored 4 students, providing feedback and advice on graduate school applications

Robotics, Science and Systems (RSS) 2018: Student Volunteer